

Why deep learning is becoming so popular.

The neural net network concepts have been there for many years so why is it that in last few years deep learning is picking up and it is becoming so much popular. The reasons are as follows:

1. Data growth

It is the growth in the volume of data if you look at any business now a days, they have some kind of idea adoption they will be using some software to manage their business they will also be having mobile app this wasn't the case a few years back and because of that the business transaction data the volume of that data is increasing also in social media so many people are posting their content that photos their likes dislikes and so on and because of the growth in the social media data it has become possible to do many interesting things such as a twitter sentiment analysis so deep learning works better if the volume of data is higher that's the general principle

behind neural networks that you need to have a huge volume of data and since the volume of data has grown up in terms of business and personal data it makes sense to do deep learning now

number two reason

2. Hardware advancements

It is advancement in hardware. Earlier laptop had only megabyte ram or even less than that and it used to take hours to run deep learning job, nowadays same job we can run on the computer within seconds so computer hardware has advanced so much. Also, we have come up with a specialized hardware such as GPU and TPU. When you talk about GPU of course the name that comes to your mind is NVidia. GPUs are awesome when you want to run computation in parallel and deep learning is perfect for that so you can run your deep learning job on NVidia GPUs also. TPU is TensorFlow processing unit, it is something that google has started building recently and there also you can learn a deep learning job so much easily so all these GPUs, TPUs and

computer hardware advancements are making it possible for normal people where we can run our jobs either locally on computer or in the cloud.

3. Python and opensource ecosystem

In the cloud third reason is python and open-source ecosystem this is probably the most important reason previously, it is hard to write programs in C++ you need to be a software engineer but python is very easy programming language to learn even a student who is in the fifth grade can learn python so nowadays what's happening is many methods mathematicians or people from statistics background if they don't know programming they can learn python in few days and they can write deep learning programs this was not possible before C++ java these are these are the programming languages which are hard to learn and python is so easy that is making it possible that any person can come in he can he or she can learn programming in few days and can start doing uh deep learning if you have math and statistics background the other thing is the deep learning of frameworks such as PyTorch and tensor TensorFlow.

PyTorch is by Facebook TensorFlow is by google so these frameworks are something that you can install on your computer for free and then you can start writing a program which makes use of neural networks the other reason that deep learning is

4. Cloud and AI boom

Advancing is uh because of cloud nowadays if you don't have GPU or TPU you can rent a server on cloud and you can run deep learning job on the cloud so you don't need to invest in um hardware right you because if you buy less GPU TPU or a heavy computer it you might have to spend a lot of money but nowadays if you want to play with it just rent a computer on cloud and you can run it there so the entry barrier overall is becoming less and less both in terms of cost as well as uh in terms of learning curve and nowadays we are living in an AI boom where every company every business wants to do artificial intelligence and AI google announced that they are going from mobile first to AI first approach and every big companies are not following their footsteps if you are not in machine learning or AI you are kind of running behind so that's what

uh majority of uh the business executives they want to invest money into artificial intelligence and machine Learning and that further accelerates the growth of deep learning so these are some of the reasons why deep learning is taking off in the next video we are going to look into what is neural network so I will use a simple analogy and explain you what exactly is neural network